Wheeled Mobile Robot Autonomy Under Degradation: A Survey of Methods and Evaluation

***Abstract*—Wheeled mobile robots operate under changing sensing, terrain, map, communication, and computational conditions that violate the assumptions of individual navigation algorithms. Reliable navigation requires estimation, planning, and control to account for these changes jointly, since errors and delays in one component affect decisions elsewhere in the system. This survey compares 229 publications from 2021 to 2026 on localization and mapping, traversability and motion control, semantic navigation, multi-robot coordination, and fault management, organizing them by the navigation assumption that each degradation violates and the information or action that each response changes. Selective sensing and fusion, adaptive motion models, map revision, and constrained action selection provide complementary responses to degradation, whose effectiveness depends on the validity of state estimates, the freshness of information, and the availability of feasible actions. Evaluation practice is compared through the measurement level, the conditions of degradation tests, and the events and assumptions underlying risk or diagnostic claims. The comparison identifies four interfaces that require joint evaluation: localization-triggered recovery, semantic reasoning with fallback under resource limits, terrain decisions conditioned on pose uncertainty, and shared-state acceptance after communication loss and reconnection. Evaluating these interfaces requires linking degradation onset, detection, and executed responses to safety and task performance under matched operating conditions. Further challenges concern transfer across platforms, computational resources, sustained operation, and safety assurance under changing information quality.**

***Index Terms*—Wheeled mobile robots, robust navigation, sensor fusion, multi-robot systems.**

I. Introduction

AUTONOMOUS wheeled robots transport goods, inspect facilities, explore unfamiliar sites, and assist people indoors and outdoors. To navigate, a robot must estimate its pose from imperfect measurements, represent obstacles and traversable space, plan a route that its chassis can execute, and track that route as new observations arrive, while a team member must also reconcile its own observations with information received from teammates. Reliable operation therefore depends both on the individual algorithms and on the information that passes between estimation, mapping, planning, and control.

Degradation occurs when operating conditions depart from the assumptions of these algorithms. Cameras encounter poor texture or illumination, laser scans lack geometric variation, wheels slip, and moved furniture invalidates a stored map, whereas semantic predictions and delayed messages introduce errors into task interpretation and coordination. Ground-robot datasets document combinations of darkness, lift transitions, abrupt braking, and uneven terrain [1], [2], and long-term deployments add dependencies on charging, network availability, and operator assistance [3]. Addressing these conditions requires methods that relate the quality of the available information to the actions that a robot can still execute safely.

Geometric, learned, and language-based methods address these requirements in combination. A semantic navigator may still rely on geometric localization and a local collision controller, and a learned planner may retain a model-based feasibility test. Representative simultaneous localization and mapping (SLAM) systems [4], [5] and the dynamic-window and timed-elastic-band planners that recur as comparators [6], [7] illustrate how these components assume different inputs and fail in different ways. Existing surveys examine individual aspects of the problem, including terrain traversability [8], learned navigation policies [9], collaborative SLAM [10], and foundation models across embodiments [11]. Table I contrasts their scopes with the system-level connections developed here.

This survey organizes the literature around three questions: which navigation assumption a degradation affects, how a method changes the information or action available to the robot, and what evidence connects that change to navigation performance. Fig. 1 summarizes the resulting organization by navigation function, response family, and open interface. The main contributions of this survey are summarized as follows. First, this survey compares responses to degradation across five navigation functions together with their sensing, model, and action requirements, distinguishing component improvements from executed recovery. Second, it examines how platform constraints, reference dependence, computation placement, and experimental design condition the interpretation of reported results. Third, it derives four interfaces between existing methods that merit joint evaluation, namely localization-triggered recovery, semantic reasoning with fallback, pose uncertainty in terrain decisions, and shared-state acceptance after reconnection.

The remainder of this survey is organized as follows. Section II introduces wheeled motion constraints, the navigation stack, and the sources of degradation, and Section III defines the scope. Sections IV–VIII develop the technical synthesis, Section IX compares evaluation practice, Section X presents open challenges, and Section XI concludes this survey.

II. Background

A. Wheeled Platforms and Motion Constraints

A wheeled mobile robot moves through driven or steered rolling contacts whose footprint, arrangement, and contact conditions constrain the paths it can execute. Differential-drive robots turn by varying the left and right wheel speeds, skid-steer robots use the same actuation with lateral slip at the contacts, Ackermann vehicles steer the front wheels with a finite minimum turning radius, and omnidirectional platforms

TABLE I
REPRESENTATIVE RELATED SURVEYS AND THE SYSTEM-LEVEL CONNECTIONS EXAMINED IN THIS REVIEW.

Review	Main organizing subject	Connection developed in this survey
Sevastopoulos and Konstantopoulos (2022) [8]	Estimation of whether terrain can be traversed	Pose uncertainty, vehicle limits, route choice, and motion outcomes
Zhu and Zhang (2021) [9]	Deep reinforcement learning for navigation	Relations between learned decisions, constraints, and deployment evidence
Lajoie et al. (2022) [10]	Collaborative localization and mapping	Shared-state validity during disconnection, local operation, and reconnection
Firoozi et al. (2025) [11]	Pretrained models across perception, decision-making, and control	Query timing, local constraints, and fallback in wheeled navigation

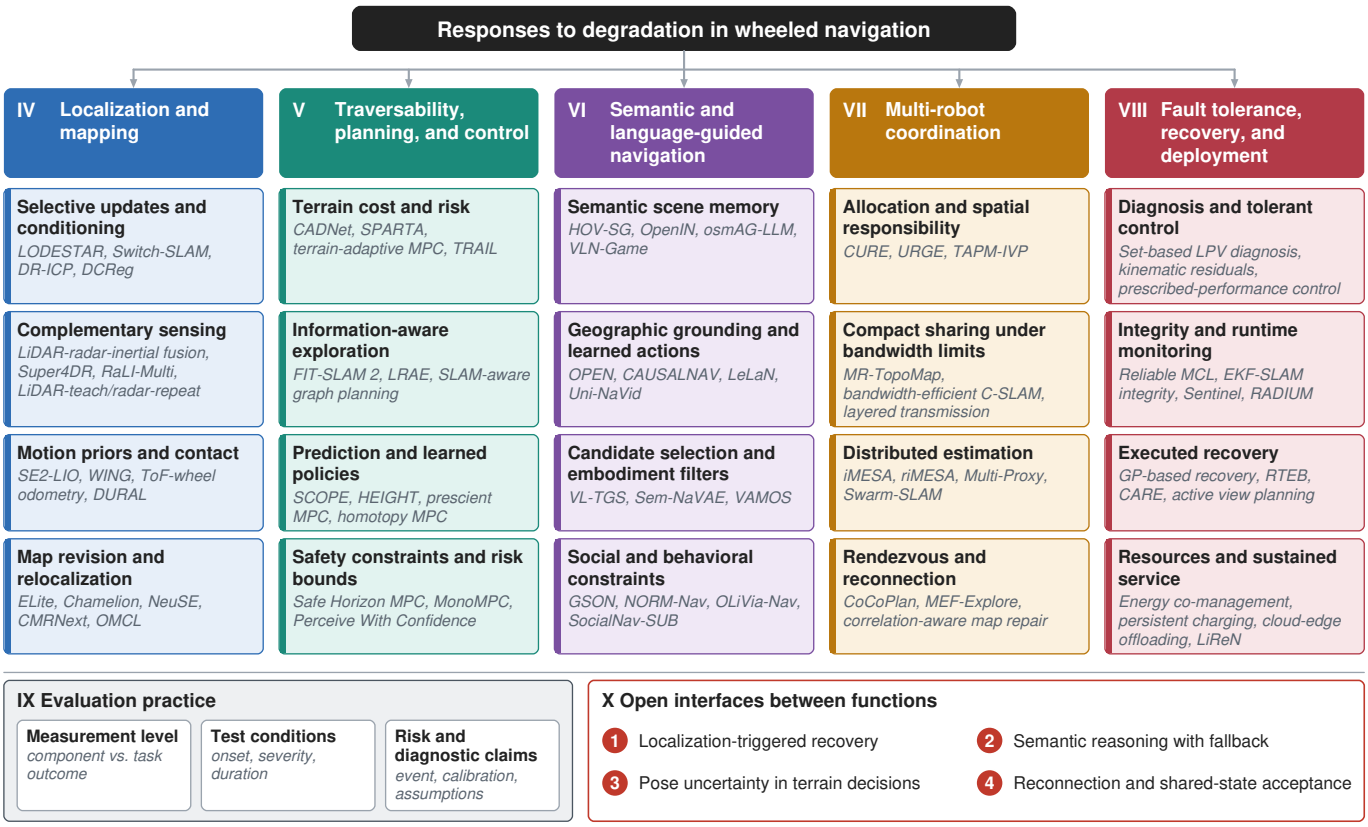


Fig. 1. Organization of the survey. Sections IV–VIII group responses to degradation by navigation function, with representative methods listed under each response family. Section IX compares evaluation practice along three axes, and Section X examines four interfaces between functions whose behavior under degradation remains insufficiently characterized.

generate lateral motion through their wheel arrangement. For a planar reference point with position (x, y) and heading ψ, a common idealization is

$$\dot{x} = v\cos\psi, \qquad \dot{y} = v\sin\psi, \qquad \dot{\psi} = \omega, \tag{1}$$

where v is longitudinal speed and ω is angular speed. Under pure rolling, a differential-drive robot with wheel radius r, track width b, and right and left wheel angular speeds ω_R and ω_L satisfies

$$v = \frac{r}{2}(\omega_R + \omega_L), \qquad \omega = \frac{r}{b}(\omega_R - \omega_L). \tag{2}$$

An ideal bicycle approximation for Ackermann steering instead gives

$$\omega = \frac{v}{L}\tan\delta, \tag{3}$$

where L is the wheelbase and δ is the steering angle. Slip violates the pure-rolling assumption, wheel calibration affects odometry, and steering and acceleration limits constrain tracking. On uneven ground, suspension, pitch, roll, clearance, traction, and payload further affect feasibility, which means that a planar collision-free path may be physically unsuitable even when localization is accurate.

B. The Navigation Stack and Its Degradation

Fig. 2 shows the principal functions of a navigation system, the points at which degradation enters, and the interfaces examined in Section X. Cameras, color and depth (RGB-D) cameras, LiDAR, radar, inertial measurement units (IMUs), wheel encoders, global navigation satellite system (GNSS) receivers, and ultra-wideband (UWB) ranging supply measurements with

calibration parameters and timestamps. The estimator infers robot state from these measurements and motion inputs, the mapping algorithm estimates spatial structure, the planner selects a route or trajectory, and the controller converts that reference into actuator commands. A supervisor can interrupt execution when measurements, models, or resources no longer support the current action. Deployed systems realize these functions through interacting maps, planners, controllers, and recovery behaviors, for example in a robot operating system (ROS) navigation stack with planner fallback [12] or under behavior-tree supervision [13].

An estimator can be described through the posterior

$$p(\mathbf{x}_t, \mathcal{M} \mid \mathbf{z}_{1:t}, \mathbf{u}_{1:t}), \tag{4}$$

where $\mathbf{x}_t$ is the robot state, $\mathcal{M}$ is the map, and $\mathbf{z}_{1:t}$ and $\mathbf{u}_{1:t}$ denote observation and motion-input sequences available through time t. Localization estimates state relative to a given reference, whereas SLAM estimates state and map jointly, commonly through visual–inertial odometry (VIO) or LiDAR–inertial odometry (LIO). ORB-SLAM3 [4] and FAST-LIO2 [5] are representative visual and LiDAR–inertial systems on which several reviewed methods build. Loop closures constrain revisited places, and an incorrect association can introduce correlated errors into many states. Downstream decisions may therefore require the covariance, association confidence, and age of an estimate as well as its mean. Geometric, topological, and semantic representations retain different information, and a representation sufficient for one decision can be insufficient for another. A place graph may support a long route while omitting a small obstacle, and a terrain label may describe appearance without establishing traction for a loaded vehicle.

In local planning, the dynamic window approach (DWA) searches admissible velocity commands, timed-elastic-band (TEB) optimization adjusts a sequence of poses and timing variables, and model predictive control (MPC) repeatedly solves a finite-horizon control problem. DWA and TEB remain common comparators in recent evaluations [6], [7]. A controller using a control barrier function (CBF) constrains inputs to keep the state within an admissible set under stated assumptions [14], [15]. For a control-affine model $\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\mathbf{u}$, a common condition is

$$\nabla h(\mathbf{x})^{\mathsf{T}} \big[f(\mathbf{x}) + g(\mathbf{x})\mathbf{u} \big] + \alpha(h(\mathbf{x})) \geq 0, \tag{5}$$

where f denotes the drift dynamics, g maps the control input $\mathbf{u}$ to the state derivative, $h(\mathbf{x}) \geq 0$ defines the admissible set, and α is a suitable increasing function with $\alpha(0) = 0$. The resulting safety argument depends on the state information, dynamics, admissible inputs, and update implementation, and penalizing a barrier violation in an objective does not establish the condition during execution.

Here, *degradation* denotes an operating change that reduces the quality or validity of the information, models, resources, or physical capabilities required for navigation. Fig. 2 distinguishes four sources that can occur within one episode. Sensing and estimation degradation includes weak texture, illumination change, poorly constrained scan geometry, GNSS denial, timing errors, and incorrect associations, and many measurements may still carry little information along one motion direction. Terrain and mobility degradation includes slopes, vegetation, soft ground, slip, and changed actuator behavior, where similar geometry can hide different friction or sinkage. Environment and map change includes moving people, altered layouts, and construction, which requires the robot to separate transient obstacles from lasting map updates. Communication and computation degradation includes packet loss, partitions, stale shared states, delayed model responses, and missed deadlines, and a restored link raises the further problem of reconciling states accumulated during the outage.

C. Accuracy, Task Performance, and Integrity

Absolute trajectory error (ATE) measures differences between estimated and reference poses after a declared alignment, relative pose error (RPE) measures motion error over a declared interval, and both are commonly summarized by the root mean square error (RMSE). Their values depend on alignment freedoms, scale treatment, synchronization, aggregation, and the dependence of the reference on the evaluated sensors. Task measures include goal success, collisions, interventions, completion time, coverage, and energy, while success weighted by path length (SPL) combines goal success with path efficiency. Neither task success nor low ATE identifies the source of a failure or the ability to recover after an alarm.

Integrity concerns the risk of an unacceptable estimation error without an adequate warning. A useful conceptual statement is

$$\Pr(E_t > A_t \ \wedge \ \text{no timely alert at } t)\,, \tag{6}$$

where E_t is an error magnitude and A_t is an application-specific alert limit. A protection level is an error bound associated with a specified integrity model, and a covariance alone is not a calibrated protection level. Particle-filter and SLAM integrity analyses account for fault hypotheses and association errors when relating alarm conditions to hazardous error [16], [17]. For navigation, the warning must also reach a response policy early enough for the motion and stopping constraints of the robot.

D. Datasets, Simulators, and Transfer

Table II relates representative resources to the evidence they can support. M2DGR and M2UD provide ground-robot sequences with sensing and motion difficulties and real-time kinematic (RTK) references [1], [2], GEODE and HK-MEMS extend the comparison to varied degeneracy and sensor conditions [18], [19], and robot/network simulation supports repeated reception and control conditions [20]. Recorded data support reproducible estimator comparisons, although replay does not change future measurements when a planner acts differently. A closed-loop simulator supplies that feedback under its model, whereas hardware adds physical interaction and execution timing, and the three settings answer different causal questions.

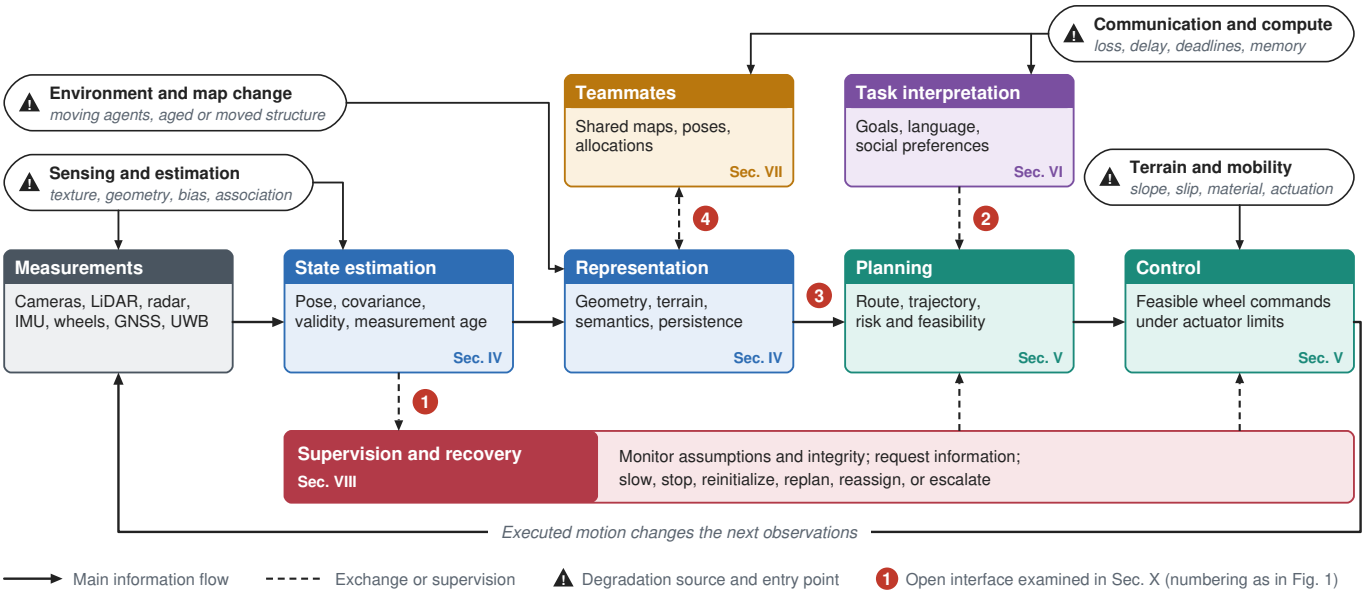


Fig. 2. Navigation functions, degradation entry points, and open interfaces. Each function is colored by the section that reviews it. Teammates and task interpretation supply information to the motion stack rather than bypassing vehicle constraints, and the numbered markers locate the four interfaces of Section X.

TABLE II
REPRESENTATIVE EVALUATION RESOURCES, THEIR INTENDED USES, AND RELEVANT LIMITATIONS.

Resource	Evaluation use	Relevant limitation
M2DGR [1]	Ground-robot localization across everyday sensing and motion difficulties	Estimation errors must be connected separately to task outcomes
M2UD [2]	Uneven-terrain localization/mapping and runtime-aware comparison	References mix RTK and offline SLAM, external checks cover selected routes, and mapping includes handheld capture
GEODE [18]	Degeneracy comparisons across sensor conditions	Sensor-specific coverage and severity affect comparisons, and replay does not test action-dependent recovery
HK-MEMS [19]	Multi-sensor localization and reference comparisons	Missing reference intervals and failed trajectories require explicit coverage reporting
Robot/network simulation [20]	Designed reception and control conditions with repeated trials	A network-in-the-loop simulation is distinct from a distributed physical deployment

III. SCOPE AND LITERATURE SELECTION

This survey covers localization and mapping, traversability and motion control, semantic navigation, multi-robot coordination, and fault management for wheeled mobile robots operating under degraded sensing, terrain, maps, communication, or computation. It includes journal articles and main-conference papers published from 2021 to 2026 that propose, evaluate, or benchmark a mechanism relevant to these functions. Studies on legged, aerial, or road vehicles are included when their mechanism transfers to wheeled navigation and are marked as such in the tables. Journal searches, consultation of RSS and CoRL proceedings, and citation follow-up yielded 229 publications by November 3, 2026. The selection is purposive rather than exhaustive, with screening assisted by a language model under preset topic allocations. Coverage of 2026 and of conference proceedings is incomplete. Consequently, the corpus supports technical comparisons rather than estimates of topic prevalence. Classical planners and controllers, such as dynamic-window planning, timed elastic bands, and barrier-function control, are discussed through the recent studies that apply them. The supplementary material documents the search sources, selection stages, and assessment depth.

Each method is compared through the navigation assumption that a degradation violates, the information or action that the response changes, and the evidence that links the response to navigation outcomes. Reported metrics are interpreted within the platform and protocol of each study, without pooling effect sizes across heterogeneous datasets or trial units.

IV. LOCALIZATION AND MAPPING

Localization and mapping methods respond to degradation through selective measurement updates, complementary sensing, motion priors, relocalization, and map revision. These responses differ in the information they retain and in their assumptions about geometry, contact, and correlations between estimates. Table III summarizes representative sensing, implementation, and evaluation settings.

TABLE III
LOCALIZATION AND MAPPING: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	**Mechanism and inputs**	**Evaluation setting** *Evaluated quantities*	**Platform** *Computation*
Uncertainty-aware localization [21]	Uncertainty-weighted prior-map matching with LIO and gated GNSS. *Inputs:* 3D LiDAR, IMU, GNSS	Acquired data *Pose error and covariance estimates*	wheeled (drive unspecified) *Unspecified*
LODESTAR [22]†	Selective historical anchoring and point selection under degeneracy. *Inputs:* 3D LiDAR, IMU	Replay *Absolute pose error and runtime*	mixed *Desktop evaluation*
ToF-Wheel [23]	Directional process-covariance inflation from geometric corrections. *Inputs:* time-of-flight camera and wheels; LiDAR replaces depth in replay	Acquired data, Replay *Trajectory error across covariance variants*	differential drive *Onboard*
DURAL [24]	Region- and degeneracy-dependent sensor fusion. *Inputs:* LiDAR, IMU, UWB, wheel encoders	Acquired data, Sim. *Localization error across fusion modes*	wheeled (drive unspecified) *Unspecified*
MM-LINS [25]†	Map hibernation, reinitialization, and pose-graph registration. *Inputs:* LiDAR/IMU from handheld and vehicle rigs	Acquired data, Replay *Trajectory error and map registration*	road vehicle *Unspecified*
SE2-LIO [26]	Near-planar pose with stochastic out-of-plane perturbations. *Inputs:* 3D LiDAR, IMU	Online estimate, Acquired data *Pose error; slope-model boundary*	wheeled (drive unspecified) *Onboard*
ProbPer-LiLo [27]†	Semantic-cluster persistence estimation and map filtering. *Inputs:* 3D LiDAR, semantic labels	Acquired data, Replay *Registration and map persistence*	mixed *Unspecified*
DMSA [28]†	Joint grid-distribution fitting over multiple scans. *Inputs:* 3D LiDAR, IMU	Acquired data, Replay *Pose error, runtime, sensor compatibility*	mixed *Desktop evaluation*
Switch-SLAM [29]†	Degeneracy-triggered introduction of visual constraints. *Inputs:* LiDAR, camera, IMU	Replay, Sim. *Trajectory error under modality failure*	mixed *Desktop evaluation*
M2UD [2]	Uneven-terrain localization and mapping benchmark. *Inputs:* LiDAR, RGB-D, IMU, GNSS; mixed reference systems	Dataset, Acquired data *Localization/mapping error and runtime*	wheeled (drive unspecified) *Unspecified*
DR-ICP [30]†	Directional feature selection and weighting. *Inputs:* LiDAR; capture IMU not integrated	Acquired data, Replay, Sim. *Tunnel pose error; pure-LiDAR scope*	mixed *Desktop evaluation*
Decoupled registration [31]†	Subspace decoupling and directional preconditioning. *Inputs:* LiDAR registration constraints	Acquired data, Replay, Numerical *Registration; missing-information boundary*	mixed *Desktop evaluation*
Semantic neural processes [32]†	Semantic neural-process elevation and uncertainty. *Inputs:* LiDAR, image semantics, poses	Acquired data *Elevation error and uncertainty; calibration needed*	road vehicle *Desktop evaluation*

† Road-vehicle, non-wheeled, handheld, or mixed-platform evaluations.

A. Localizability and selective measurement updates

Geometric degeneracy occurs when measurements constrain some pose directions much less than others. LODESTAR retains information in the weak directions by selecting historical Schmidt–Kalman anchors from the condition of the LiDAR measurement Jacobian and by choosing points according to their contribution along those directions [22]. With a common FAST-LIO2 backbone, absolute pose error (APE) on the SubT-MRS Long Corridor recording falls from 3.113 to 1.386 m, while average desktop computation rises from 21.04 to 28.69 ms. The ordering is not uniform, since on Short exp the APE is 0.429 m versus 0.412 m for the baseline, and the method still requires LiDAR returns. Other detectors characterize degeneracy before acting on it. A screw-theoretic analysis classifies translational, helical, and spherical degeneracy from point, line, range, and bearing constraints [33], normal histograms guide feature selection and weighting along weak tunnel directions in DR-ICP [30], and a DBSCAN-based detector adapts its decisions from accumulated condition-number history and compensates with IMU motion after a startup phase [34]. Decoupled registration preconditions weak subspaces while preserving the least-squares objective [31]. Better conditioning exploits weak but nonzero information without restoring an absent constraint or rescuing an initialization outside the convergence basin, which means that selection, compensation, and preconditioning require different controls

when their benefits are compared.

B. Complementary sensing

Complementary sensors supply information that a degraded scan lacks. Switch-SLAM introduces visual constraints when normalized Hessian eigenvalues indicate LiDAR degeneracy and falls back to inertial propagation when both modalities fail [29]. Degradation-resilient LiDAR–radar–inertial odometry keeps LiDAR constraints along observable directions while radar velocity supplies independent motion information, with flying-robot tests in obscurants [35], and an equivariant LiDAR–radar–inertial filter estimates inertial biases and rejects radar outliers under an exact-map observability assumption [36]. Radar-centric estimators must also cope with sparse, noisy returns. A continuous-trajectory radar–inertial estimator propagates pose uncertainty to the timestamp of each point and down-weights uninformative points during registration and mapping [37], whereas Super4DR learns radar odometry through self-supervision and completes the radar map with Gaussian primitives guided partly by image depth [38]. RaLI-Multi uses inter-vehicle UWB ranges to correct LiDAR–inertial drift along the detected degenerate direction, assuming at least one well-constrained horizontal direction, while a stopped anchor vehicle optimizes and broadcasts a shared pose graph [39]. MM-LINS instead hibernates an unreliable map, reinitializes, and later registers overlapping maps [25], trading continuous use of one map for recovery through later overlap. Selective updates need surviving geometry, fusion needs measurements that remain informative, and reinitialization needs later overlap, and none of these requirements follows from a degeneracy alarm alone.

C. Motion priors, contact, and adaptation

Ground geometry can enter estimation as a reweighting, a motion prior, or an explicit factor. AESF-LIO adaptively fuses separate ground and non-ground corrections without adding an independent observation [40], SE2-LIO parameterizes pose on $SE(2)$ with out-of-plane perturbations modeled as noise [26], and a sparse-channel LiDAR system adds a ground-normal factor and intensity features to suppress vertical drift, although its detector misses gradual slope changes and its indoor references share LiDAR registration with the estimator [41]. These designs address different error causes, and significant slope variation can invalidate the near-planar priors.

Wheel and inertial measurements bridge failures of external sensing when contact and excitation remain adequate. Wheel-aided VIO propagates inertial–wheel motion during visual failure [42], PL-VIWO adds motion-consistency rejection and point-assisted line triangulation evaluated on recorded vehicle data with fixed calibration [43], and a quadratic motion-manifold estimator integrates wheel measurements on a local ground model that extremely uneven farmland violates [44]. WING learns corrections for IMU bias and wheel errors caused by skidding and vertical bouncing on a continuous B-spline ground manifold, which supports interoceptive-only odometry as a fallback whose benefit depends on how closely the ground follows the manifold [45]. Contact uncertainty can also be estimated explicitly, as when a tire-model observer fuses wheel-level velocity estimates by covariance intersection and excludes inconsistent wheels [46], and time-of-flight (ToF)–wheel odometry inflates process covariance only along the remainder that differential-drive motion cannot explain [23]. DURAL switches between LiDAR–inertial–UWB estimation inside a surveyed anchor region and wheel-aided estimation outside it [24]. Learned kinematic factors [47] and the hierarchical modality reweighting of Super Odometry [48] train motion priors while perception is reliable and freeze them when supervision degrades, which limits the incorporation of unreliable measurements.

Contact and calibration provide additional observations. Lug-generated wheel traces reveal slip on suitable soil [49], and commanded collisions with known-map geometry constrain pose in a filter, although the hardware tests use manually commanded contacts [50]. GNSS-aided IMU–odometer calibration estimates wheel scale and extrinsics but leaves vertical translation unobservable under planar motion [51], and a calibration parameter should not absorb transient contact failure. IL-LIO learns point-to-voxel residuals and uncertainty for nonplanar geometry [52], although learned residual variance is not an independently calibrated integrity bound.

D. Dynamic geometry and persistent maps

Dense multi-scan adjustment jointly fits grid-cell distributions over several scans and runs slower than recording time [28]. Dynamic-scene estimators choose between excluding moving observations and estimating them. Exclusion keeps the static map clean but can remove much of the geometry in crowds, as in semantic rejection of person features in hospital RGB-D SLAM [53] and in GPU-free asynchronous mask propagation in NGD-SLAM, whose reported tracking time excludes concurrent detection [54]. DynoSAM instead estimates rigid-object motion in world-centric factor graphs, relying on full-batch optimization without loop closure [55], and PG-SLAM and JPG-SLAM retain dynamic appearance through Gaussian representations that are not real time or lack loop closure [56], [57]. Improved rendering does not by itself establish a better obstacle map, and the choice depends on whether the downstream task needs static geometry or object motion.

Across sessions, map maintenance additionally requires an estimate of how long structure persists. ProbPer-LiLo estimates semantic cluster persistence with a survival prior [27], ELite weights registration by local and cross-session ephemerality [58], and a lifelong LiDAR mapper stores positive and negative map differences that allow earlier versions to be reconstructed [59]. Chamelion learns change detection from synthesized structural changes and separates disappearance from occlusion before updating construction-site and office maps, performing best when registration error and voxel size resemble its training data [60]. For automotive radar, multi-session loop verification and a temporal occupancy grid retain persistent points for later localization under a planar-trajectory assumption [61]. NVMS-SLAM keeps normal-bearing surfaces apart to avoid merging opposite faces of thin walls and

refines alignment offline [62], and NeuSE updates object maps after layout changes within trained categories [63]. Most of these evaluations use recorded data, handheld acquisition, or injected changes, and map consistency alone does not establish how a moving robot responds to a revised map.

E. Prior-map and cross-modal localization

Localization against a compact or structural prior reduces the online representation but becomes sensitive to changes in the mapped environment. ERPoT matches sparse scans to polygon maps after ground removal [64], Area Graph localization matches filtered scans to walls and doors with WiFi and barometric initialization [65], BIM localization estimates as-designed discrepancies before weighting structural constraints [66], and orchard trunk maps suppress seasonal foliage while making global recovery depend on adequate trunk support [67]. An uncertainty-aware outdoor localizer weights NDT matching by estimated covariance and gates GNSS, which can reject accurate fixes after extended LiDAR loss [21]. Relocalization obtains new hypotheses from appearance or structure, including cross-attention image retrieval on pushed-robot recordings [68], object-level scene graphs that tolerate staged object moves [69], distilled LiDAR–camera place recognition [70], rendered point-cloud, mesh, or neural-radiance maps [71], and density-map loop closure whose maps can inherit front-end degeneracy [72]. Retrieval provides a candidate rather than a verified pose, and recovery requires subsequent pose validation.

Cross-modal matching relaxes the requirement that map and query share a sensor. CMRNext casts camera-to-LiDAR matching as optical flow followed by perspective-n-point estimation and transfers across camera setups, although it is trained on driving data, is not yet real time, and receives an initial pose for every frame [73]. OMCL scores Monte Carlo hypotheses with open-vocabulary features ray-cast from a three-dimensional map and can initialize from a language description, while requiring map-scaled odometry and a static map [74]. LiDAR Teach, Radar Repeat applies cross-modal registration in closed loop [75]. Forward-looking 4D radar is localized against LiDAR keyframes from a teaching pass using a Doppler motion prior and an alignment of radar power with LiDAR intensity, and registration errors drive label-free fine-tuning. Route repetition on a four-wheel differential-drive platform covers smoke, moving traffic, and structural change over months, while obstacle avoidance still relies on other sensors.

F. Cooperative estimates and evaluation resources

Cooperative estimators add constraints while introducing correlations. CoVOR-SLAM corrects monocular scale with inter-agent ranges in a similarity-pose graph evaluated on recorded data with one stationary rover [76], a Lie-group estimator propagates cross-covariances among wheel, inertial, and marker estimates over near-straight indoor motion [77], and a fault-tolerant prior-map estimator diagnoses observation, landmark, and odometry faults and lets a repaired landmark benefit a later collaborator [78]. Benchmarks and active sensing qualify robustness claims. M2DGR and M2UD expose combined sensing and motion difficulties [1], [2], HK-MEMS and GEODE vary degeneracy, carriers, and reference construction [18], [19], and robot-mounted lighting in LUCID shows that improved visual appearance does not necessarily improve odometry [79]. AirSLAM combines learned point and line features with map reuse and reaches 40 Hz on an embedded GPU, although it reduces to a point-feature system without structural lines [80]. Among uncertain terrain representations, semantic neural processes predict elevation with uncertainty that still requires calibration [32], E2-BKI propagates evidential semantic uncertainty through geometry-aligned kernels [81], and a multi-fidelity GP explorer weights field samples by localization uncertainty on an aquatic robot [82]. How a calibrated pose distribution should change wheel-footprint clearance or speed remains open, as discussed in Section X.

G. Synthesis

The main distinction among these methods is which information survives degradation. An estimator can retain constraints in observable directions, add independent motion information from radar, range, wheel, or contact sensing, or bridge weak external sensing with learned or geometric priors whose adaptation is useful only while supervision remains credible. Relocalization and map revision correct accumulated inconsistency through new correspondences, with further requirements for overlap, persistence, cross-modal registration, and correlation handling. Their diagnostics differ in meaning, since geometric conditioning, motion consistency, and place similarity cannot be treated as a common probability of localization failure.

V. Traversability, Motion Planning, and Control

Motion planning and control connect the estimated environment to executable behavior. Under degradation, feasibility depends on terrain support, obstacle motion, state uncertainty, actuator limits, and the age of control information. The methods below address these dependencies through traversability costs, information-aware exploration, learned prediction and policies, safety constraints, and predictive control, and Table IV summarizes representative implementations.

A. Terrain support and traversability costs

Traversability is a relation between terrain and robot capability, and the reviewed methods express it through quantities with different physical meanings. CADNet predicts heading-dependent accessible range from LiDAR history for a TEB planner on a SCOUT robot [84], and GA-Nav projects attention-based terrain segmentation into a cost map combined with elevation [85]. A risk-aware learner labels traversed footprints and adds weak obstacle labels before predicting unvisited cells, although its predictions are evaluated only on manually driven recordings [96]. PE-RRT* infers hidden support planes in vegetation with multivariate Gaussian processes and tracks the route with nonlinear MPC [97], while

TABLE IV
TRAVERSABILITY, PLANNING, AND CONTROL: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
LRAE [83]	Unknown-region safety classification and route optimization. *Inputs:* 3D LiDAR	Hardware, Sim. *Exploration time and travel cost*	skid steer *Unspecified*
CADNet [84]	Heading-dependent accessible-range prediction for TEB planning. *Inputs:* LiDAR history	Hardware, Replay, Sim. *Traversability prediction and navigation*	wheeled (drive unspecified) *Onboard*
GA-Nav [85]	Semantic terrain segmentation combined with elevation costs. *Inputs:* RGB images, 3D LiDAR	Hardware, Replay *Terrain segmentation and navigation success*	wheeled (drive unspecified) *Onboard*
Anisotropic exploration [86]	Direction-dependent quasi-static terrain costs. *Inputs:* LiDAR terrain map, FAST-LIO pose	Hardware, Sim. *Exploration, travel cost, body roll*	differential drive *Onboard*
FIT-SLAM2 [87]	Keyframe-linked frontiers and information-aware adaptive roadmaps. *Inputs:* RGB-D/LiDAR (simulation); stereo depth (hardware)	Hardware, Sim. *Exploration time, pose error, and runtime*	skid steer *Unspecified*
Vehicle–cloud MPC [88]	Switching between onboard linear and cloud nonlinear MPC. *Inputs:* externally measured poses	Hardware, Sim., Theory *Tracking and collision outcomes*	Ackermann *Mixed placement*
CECBF [89]	CBF constraints with circulation and hold logic. *Inputs:* 2D LiDAR, external robot pose	Hardware *Obstacle avoidance and escape from equilibria*	differential drive *Local offboard*
Safe Horizon MPC [90]	Scenario constraints on joint collision risk with deceleration fallback. *Inputs:* robot/pedestrian poses and velocity estimates	Hardware, Sim. *Task duration and planned collision risk*	skid steer *Unspecified*
Spatiotemporal-Tubes [91]	Adaptive spherical tubes for prescribed-time reach-avoid. *Inputs:* robot state, online unsafe-set geometry	Hardware, Sim., Numerical *Reach-avoid and prescribed-time behavior*	omnidirectional *Unspecified*
Perceive With Confidence [92]†	Environment-level calibration of obstacle inflation for constrained planning. *Inputs:* RGB-D point clouds, VIO	Hardware, Sim. *Collision-free outcomes and calibration*	legged *Onboard*
Homotopy MPC [93]	Parallel optimization of homotopy alternatives. *Inputs:* externally tracked robot/pedestrian states	Hardware, Sim. *Dynamic avoidance; external pose*	wheeled (drive unspecified) *Unspecified*
Visual-geometric MPPI [94]	Expert-trained cost, speed, and novelty maps for MPPI. *Inputs:* RGB features, LiDAR geometry	Hardware, Acquired data *Interventions, speed, and terrain response*	wheeled (drive unspecified) *Unspecified*
SPARTA [95]	Angle-conditioned risk queried during planning. *Inputs:* prepared point cloud, LiDAR/IMU pose	Hardware, Sim. *Simulated tire risk; virtual risky hardware branch*	skid steer *Onboard*

† Road-vehicle, non-wheeled, handheld, or mixed-platform evaluations.

a multilevel planner checks settling, support polygon, and body collision for a tracked robot whose wheeled variant is simulated [98]. An anisotropic explorer penalizes motion by slope and travel direction with a quasi-static Euler Cost that omits slip and suspension transients [86]. A footprint label, a support plane, a stability margin, and a cross-slope penalty are not interchangeable estimates of traversal risk. Each is also computed from an estimated pose, which means that upstream pose error can alter the planning input even when the terrain model is unchanged.

Recent planners condition terrain risk on approach direction and uncertainty and embed it in optimization. SPARTA predicts angle-conditioned risk distributions as reusable Fourier functions from a fully observed prior point cloud [95]. Its tire-deformation consequences are simulated, and its physical demonstration uses a virtual risky branch. A visual-foundation-model approach learns cost, speed, and novelty maps from expert driving for MPPI, and operator corrections expose residual failures that offline trajectory agreement conceals [94]. A terrain-adaptive ϵ-constraint MPC minimizes path length while bounding a pose-stability cost adjusted online from terrain descriptors and corrects car-like dynamics with a sparse GP [99]. Its success falls from 94% in simulation to 80% outdoors on a Scout Mini, which the authors attribute to wheel slip and sensor noise, and it assumes an offline terrain mesh. TRAIL learns an implicit field of terrain attributes from RGB and LiDAR and optimizes path geometry and speed along its gradients, with field interventions caused by wet-ground traction loss and failures attributed to occlusion and limited field of view [100]. These systems already make risk-sensitive terrain decisions, whereas map completeness, uncertainty calibration, traction, and physical consequences remain separate evaluation questions.

B. Exploration with information and geometry

Exploration chooses where to improve the map while preserving the ability to localize. FIT-SLAM 2 attaches frontiers to SLAM keyframes, adapts a traversability roadmap after pose changes, and rejects edges with low expected landmark information [87]. With a common controller and simulator pose, completion takes 880 s versus 1118 s for FAEL in Cave but 1809 s versus 1706 s in Corridor, and Fisher-information evaluation raises mean planning time from 20.8 to 63.0 ms. Confidence-rich exploration combines particle-trajectory entropy with occupancy confidence and executes the loop on hardware [101], whereas an uncertainty-frontier planner scores coverage with signed relative entropy under known orientation [102]. Other explorers change the representation of unexplored space, including safety levels for unknown regions in LRAE [83], octree category distributions for semantic exploration [103], and overlapping free polytopes in FRTree, whose body-containment corridor does not enforce wheeled curvature or rolling-contact limits [104]. A SLAM-aware planner inserts informative loop closures over a supplied topo-metric graph [105], and potential-field MPC integrates frontier and unknown-space costs on a six-wheel skid-steer robot without systematic baselines [106]. Observable routes can therefore cost travel and computation, and the benefit of estimation-aware planning depends on prior availability and useful loop opportunities.

C. Learning, prediction, and interaction

Learning enters planning at different points. NAMR-RRT learns where a risk-aware tree samples [107], prescient navigation passes multimodal pedestrian hypotheses to nonlinear MPC that may exhaust its time budget [7], and a learned relaxation of optimal reciprocal collision avoidance (ORCA) predicts escape direction and avoidance responsibility [108]. SCOPE predicts distributions of future occupancy grids from robot motion, tracked objects, and a local static map and inserts their mean and uncertainty into the costmaps of existing planners, with distillation enabling embedded inference [109]. Its navigation comparisons are mainly simulated, and its constant-velocity robot model fails during abrupt rotation. Homotopy-driven MPC optimizes distinct trajectory classes in parallel from externally tracked poses [93]. Candidate diversity reduces local inconsistency without compensating for a missed pedestrian or an incorrect interaction model.

Learned local policies encode observation history implicitly. Visual proximal policy optimization (PPO) adds a motion-perceptibility auxiliary task [110], depth-based recurrent PPO is compared with DWA and TEB in simulation [6], and a LiDAR actor–critic controller uses spatiotemporal attention under adaptive Monte Carlo localization (AMCL) on a prebuilt map [111]. HEIGHT separates pedestrians from static obstacles in a heterogeneous spatio-temporal graph and transfers from a low-fidelity simulator to four everyday environments, although its deployed obstacle input assumes a map and reliable localization [112]. A rough-terrain sim-to-real policy trained with terrain scaling and pose noise still fails under vibration and inaccurate reconstructed geometry, with offboard inference over Wi-Fi [113], and lifelong navigation can learn only where an initial DWA policy supplies useful experience [114]. As the learned-navigation review notes [9], the training data and observation representation determine which disturbances such controllers can accommodate, and recurrence alone does not establish robustness to localization failure.

D. Safety filters and explicit risk conditions

A safety filter modifies a nominal command when a constraint would otherwise be violated. A LiDAR-based CBF controller fits a GP to current obstacle edges and uses its mean and gradient in a quadratic program, with an invariance argument that requires known state, unconstrained inputs, and suitable updates while excluding GP variance [15]. A circulation-embedded CBF adds a tangential constraint and hold logic [89], dynamic movement primitives receive a CBF correction to their forcing term [115], an adaptive high-order CBF filters iterative LQR commands under input bounds [14], a disturbance-estimation CBF compensates matched uncertainty on a tracked robot [116], and safe reinforcement learning learns state-dependent high-order barrier penalties alongside soft actor–critic [117]. Their evidence ranges from external pose and offboard computation to numerical studies, and high-order tuning affects both reaction time and feasibility under actuator bounds.

Other controllers constrain an explicit risk event. A distributionally robust CBF uses noisy barrier samples [118], a conditional value at risk (CVaR) controller selects among feasible risk levels in crowds [119], and a risk-aware-zone barrier expands an avoidance region under input saturation with a score that is not a calibrated probability [120]. Safe Horizon MPC samples obstacle trajectories, applies a plan only after slack and support checks, and otherwise decelerates [90]. In 100 Gaussian eight-pedestrian simulations with a joint-risk specification of 0.05, its mean traversal takes 16.0 s (standard deviation 1.9 s) versus 18.4 s (2.4 s) for chance-constrained MPC, with maximum estimated plan collision probabilities of 0.0092 and 0.0019. Faster traversal thus accompanies a less conservative plan under the assumed distribution, and these probabilities are not empirical collision rates. Perceive With Confidence calibrates obstacle inflation over sampled environments for a planner with an inevitable-collision-set constraint on a legged robot [92], and conformal crowd navigation limits cumulative intrusion into calibrated pedestrian regions on a Mecanum robot with offboard computation [121]. MonoMPC learns the distribution of minimum clearance for candidate controls from monocular depth and minimizes a distribution-based risk [122]. Jackal trials report up to seven times fewer collisions than a navigation stack on the same estimated depth, with planning on a networked laptop and no temporal memory. Chance constraints, CVaR, conformal coverage, and calibrated clearance concern different events, which comparisons must preserve rather than treat every confidence value as a collision probability.

E. Tracking, mission costs, and network support

Tracking control shows how model error, command age, and mission utility enter execution. A GP-MPC controller learns residual dynamics offline from field data and uses only the GP mean [123], and predictive learning refines repetitive Mecanum trajectories under external UWB pose [124]. Delay-aware MPC predicts current states from delayed packets and buffers future controls [125], whereas vehicle–cloud MPC runs onboard and remote optimizers in parallel and stops on timeout [88]. Buffered prediction and fail-safe stopping are different responses to communication loss. Mission-dependent costs add another reason to change a route, as in infrastructure-based hazard inference added to a cost map [126] and in payload- or sensing-aware MPPI on a compressed scouting map [127]. Prescribed-time spatiotemporal tubes assume bounded disturbances and terminal separation without explicit input constraints [91], memory-event-triggered MPC reduces repeated solves for a no-slip Ackermann model in hardware-in-the-loop tests [128], curvature-constrained vector fields co-design motion and saturation on an Ackermann vehicle with motion capture [129], and universal constrained tracking filters references that exceed prescribed state bounds [130]. Neither a reach-avoid proof without input limits nor fewer solves establishes timely, feasible control on a saturated onboard system.

F. Synthesis

Motion methods differ in what they change: the terrain cost, the sampled route, the predicted occupancy or obstacle motion, or the final command. A terrain cost can favor better support without enforcing stability, and a learned policy can avoid familiar interactions without identifying a feasible response to an unseen fault. Explicit constraints make admissibility testable only under their state, dynamics, and update assumptions. The resulting trade-offs concern progress as well as conservatism, and comparisons should establish which change produces the reported task benefit.

VI. Semantic and Language-Guided Navigation

Semantic navigation combines task interpretation, scene memory, and motion generation. A semantic model may output a room, an object instance, frontier scores, a path, a cost layer, or an action sequence, and this output determines how a reasoning error propagates into planning and control. Table V summarizes representative sensing and evaluation settings.

A. Semantic memory and scene structure

Persistent semantic maps support relational queries but must be updated when objects move. HOV-SG organizes a mapped building into floors, rooms, and objects for hierarchical retrieval [135], DualMap searches stable anchors globally before checking objects locally [143], and OpenIN updates carrier–object existence confidence when an expected object is not observed [133]. These updates must distinguish removal from occlusion or poor depth. osmAG-LLM assumes from the outset that objects move. It keeps a lightweight text-based area graph for context, asks a language model for likely locations, and confirms targets by online detection [144], with 27 of 30 building trials reaching a correct position and detection errors causing most failures. Goal-conditioned maps store less reusable structure. VLFM projects image–language similarity into a frontier value map until a detector supplies the goal [134], and VLN-Game builds an object-centric three-dimensional map and uses a game-theoretic vision-language model (VLM) to disambiguate attribute-rich descriptions, with quantitative results from Habitat-Matterport scenes [145]. SysNav reserves VLM calls for room-level decisions and leaves viewpoint selection and waypoint execution to geometric modules [146]. Multimodal spatial language maps add image and audio features to an offline map, although a remembered sound marks where it was heard rather than its source [140], and temporal scene-object graphs report corner trapping and premature termination among their physical failures [147]. Co-NavGPT extends semantic reasoning to team frontier assignment executed by a geometric policy, with perception and planning on a workstation [132].

B. Geographic grounding, priors, and learned actions

Outdoor instructions often refer to places beyond the sensor horizon. OPEN associates visual semantics with OpenStreetMap and replans locally when road geometry differs from the map [148], G-DRAGON constrains generated identifiers to existing map entities before composing a behavior tree [149], and CAUSALNAV retrieves destinations from an object–building–cluster graph and combines global routing with constrained local control [150]. Their requirements differ in prior exploration, map coverage, heading initialization, and remote computation. Human and recorded priors raise similar coverage questions. HAM-Nav associates live landmarks with an imperfect hand-drawn map and predicts omitted ones before A*/TEB execution [151], and Mobility VLA retrieves a goal frame from a demonstration tour and routes over an offline graph without exploring [152].

Learned policies map language and images directly to motion. LeLaN trains a last-mile policy from labeled robot and action-free video [137]. In 150 trials with 28 initially visible objects, its success within 0.2 m is 0.89 versus 0.67 for OWL-v2 with ZoeDepth, while a separate 15-trial ablation reaches 0.60 arrival with collision training and 0.13 without it. These results separate target grounding from obstacle handling without establishing open-world search or a collision-risk bound. Uni-NaVid compresses multiscale video history before predicting discrete primitives whose vocabulary bounds the available actions [136], and LGX prompts a language model for exploration directions with random exploration after an invalid response [153].

C. Candidate selection and behavioral constraints

Selecting among explicit candidates confines semantic errors to a choice among feasible alternatives. VL-TGS generates LiDAR-based trajectories, projects them into an image for GPT-4V ranking, and follows the choice with DWA while later decisions are pending [138]. Its ranking cannot recover a

TABLE V
SEMANTIC AND LANGUAGE-GUIDED NAVIGATION: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
VLM-Social-Nav [131]	Cue-triggered VLM judgments used as social DWA costs. *Inputs:* RGB images, LiDAR	Hardware *Task success, collisions, and social behavior*	differential drive *Mixed placement*
Co-NavGPT [132]†	VLM frontier allocation with geometric goal execution. *Inputs:* RGB-D, shared map, robot states	Hardware, Sim. *HM3D search; separate hardware demos*	legged *Onboard pose; offboard perception/planning*
OpenIN [133]	Carrier–object memory with visibility-dependent updates. *Inputs:* RGB-D, language/image queries, SLAM pose	Hardware, Sim. *Success across sequential object queries*	wheeled (drive unspecified) *Unspecified*
VLFM [134]†	Image–language similarity projected into a frontier value map. *Inputs:* RGB-D, odometry	Hardware, Sim. *Goal success and path efficiency*	legged *Unspecified*
HOV-SG [135]†	Hierarchical retrieval over floor, room, and object graphs. *Inputs:* RGB-D, LiDAR-derived poses	Hardware, Sim., Replay *Object retrieval and navigation success*	legged *Unspecified*
Uni-NaVid [136]†	Multiscale video memory and discrete action prediction. *Inputs:* monocular RGB video, instruction	Hardware, Sim. *Goal success and path efficiency*	legged *Local offboard*
LeLaN [137]	Receding-horizon language-conditioned velocity prediction. *Inputs:* RGB image, text instruction	Hardware, Replay *Target arrival and action-generation time*	differential drive *Onboard*
VL-TGS [138]	VLM selection among generated trajectories with local DWA control. *Inputs:* LiDAR, RGB, velocity, GPS	Hardware, Acquired data *Outdoor traversal and model-response time*	wheeled (drive unspecified) *Mixed placement*
GSON [139]	Group-aware routing with predictive barrier-constrained control. *Inputs:* 2D LiDAR, RGB images	Hardware, Replay *Social disturbance, travel time, and jerk*	differential drive *Unspecified*
AVLMaps [140]	Multimodal spatial retrieval and goal APIs. *Inputs:* offline RGB-D/pose/audio map, instructions	Hardware, Sim. *Multimodal goals; prepared-map validity*	wheeled (drive unspecified) *Unspecified*
VLM-GroNav [141]	Interaction-grounded terrain costs and DWA. *Inputs:* RGB, LiDAR, proprioception, GPS, aerial images	Hardware *Outdoor task outcomes; GPS/cloud dependencies*	skid steer *Mixed placement*
SocialNav-SUB [142]	Social scene-understanding VQA benchmark. *Inputs:* recorded images, pedestrian annotations	Dataset *Human/rule agreement; no closed-loop control*	wheeled (drive unspecified) *Mixed placement*

† Road-vehicle, non-wheeled, handheld, or mixed-platform evaluations.

route that the generator never proposes. Sem-NaVAE applies the same generate-and-select structure without a map, scoring conditional-VAE proposals with lightweight open-vocabulary segmentation and reporting 90% success on 120–240 m campus and park routes with few repetitions per route [154]. VAMOS adds an embodiment filter, in which a per-robot affordance model trained in simulation re-ranks the image-space paths proposed by a VLM planner [155]. With the planner unchanged, the affordance model of a wheeled robot raises its success on a ramp-or-stairs task from 60% to 90%, which shows that semantic proposals must be filtered by the capabilities of the executing chassis.

Social and behavioral constraints separate preference from collision geometry. VLM-Social-Nav converts cue-triggered judgments into social costs for DWA [131], VLM-informed path selection ranks clustered detours before an ORCA controller computes velocities [156], and GSON supplies group labels to a mid-level planner above a barrier-constrained predictive controller [139]. NORM-Nav parses free-form instructions with a language model and writes semantic, directional, and speed layers into costmaps for standard grid planners, with quantitative comparisons in simulation [157]. OLiVia-Nav distills large-model social reasoning into lightweight encoders that are updated online and is evaluated against expert trajectories and personal-space violation rather than collision rates [158]. Safe-VLN masks predicted waypoints with simulated occupancy and reselects after collision [159]. SocialNav-SUB tests the reasoning premise directly, showing that strong VLMs still trail human consensus and position-based rules on social scene questions [142]. Physical feedback can ground semantic terrain estimates, as VLM-GroNav revises terrain costs and waypoints from visual and proprioceptive exemplars while depending on GPS and a cloud API [141]. A grouping or grounding error can thus alter the route even when the local controller avoids detected people, whereas an occluded pedestrian can invalidate the controller itself.

D. Memory, timing, and embodied execution

Semantic decisions remain valid for different durations. TARIC triangulates intermittent goal cues into a particle

memory whose dispersion limits heading deviation during cue-free travel [160]. SysNav queries at room transitions, GSON uses updated keyframes, and the social path selector associates asynchronous responses with current candidates [139], [146], [156], whereas VAMOS plans at 1 Hz on a laptop GPU [155] and OLiVia-Nav moves reasoning into compact encoders [158]. A room decision can outlive a pedestrian detour, which means that response age must be judged against the referenced scene, and a fast local controller can absorb a slow query only while it retains a valid action. Embodiment then determines how a semantic choice becomes motion. VLFM uses different low-level controllers in simulation and on Spot [134], Uni-NaVid runs remote inference on a quadruped [136], and LeLaN executes on a wheeled base [137]. For wheels, turning in place, lateral motion, and reversing are available only if the chassis supports them. Map or tour preparation, cloud availability, and the actions available during pending inference also belong in the comparison [140], [151], [152].

E. Synthesis

The reviewed systems connect semantic reasoning to motion through scene memory, geographic grounding, learned actions, cost layers, or candidate selection. Persistent maps support reuse but require validity updates, direct policies depend on their history and action vocabulary, and candidate-based systems remain limited by the generator, the embodiment filter, and the local controller. Semantic correctness and collision avoidance therefore require separate evaluation, and matched trials with incorrect or delayed inference can establish how rejection and fallback affect completion, collisions, and resource use.

VII. Multi-Robot Coordination

Multi-robot coordination involves task allocation, information exchange, state fusion, and motion interaction. Teams exchange frontiers, maps, observations, poses, or position histories, and under communication degradation local operation and shared-state consistency become separate objectives. Table VI compares representative mechanisms and evaluation settings.

A. Task allocation and spatial responsibility

Allocation reduces redundant work through a common state. Centralized planners partition merged maps into dynamic Voronoi cells in CURE [172], assign known targets over a sparse Delaunay graph with priority yielding in DMT-CPP [173], or assign tasks before the mission while penalizing capability shortfall [174]. Distributed methods reduce negotiation without removing shared state. CQLite exchanges selected Q-values and frontiers [175], LPFE predicts teammates' frontier choices from a common map and last known positions [176], and 4CNet predicts map structure from shared trajectories with an estimated confidence [177]. Such inference avoids messages only while robots hold compatible inputs. URGE abstracts the map into subregions with exploration states and allocates them by jointly minimizing path cost and balancing load [178]. Its full-set exchange with an explored-overrides-unexplored rule lets regional maps converge after a link returns. Its baseline comparisons assume ideal communication in simulation, whereas three-robot runs in a forest, a parking lot, and a tunnel use an ad hoc mesh network. Terrain-aware partitioning (TAPM-IVP) extends spatial responsibility to uneven ground with communication-radius-constrained regions and traversability-aware subgoals [166]. In ten simulated trials of Scene 1, each robot succeeds in 90% of trials, compared with 40% and 30% for an ablation that removes both partitioning and greedy subgoal selection. These rates are per robot, and the plane-fitting covariance describes terrain scatter rather than propagated pose uncertainty.

B. Compact sharing and distributed estimation

Compact representations trade bandwidth against what teammates can infer. MR-TopoMap shares topological vertices with place descriptors while keeping local occupancy for avoidance [179], Cross-rank exchanges position histories and assigns directions at intersections [168], and WiSER-X infers relative positions from radio pings that require an interpretation when they are missing [180]. SlideSLAM buffers exchanged object observations until a loop closure fixes the frame transform [181], and Commerge minimizes transmitted scans for central submap registration without adapting to radio capacity [164]. Bandwidth-efficient collaborative SLAM separates lossy mutual-information keyframe selection from lossless entropy coding under a central back end [171]. Feedback-guided hierarchical transmission sends a pose-and-factor skeleton, prioritized geometry, and semantic labels in layers and reallocates bandwidth each round from back-end loop-closure and risk signals [182]. Its dataset-replay evaluation reports estimated rather than measured collision risk, and the authors note that round-wise guidance can become stale under asynchronous links. PROMO-Orchard gates submap merging with orchard priors and posterior checks against a reference built by the same backbone [183].

Distributed estimators must also handle delayed and repeated information. iMESA keeps local incremental factor graphs and updates consensus variables through a two-stage exchange that must complete once started [165]. riMESA, a direct extension by the same authors, removes this blocking requirement [184]. It formulates collaborative SLAM as edge-based consensus ADMM, rejects outliers with a robust local solver, and discards failed exchanges, with experiments that replay recorded LiDAR factors under Wi-Fi or radio connectivity models. Multi-Proxy combines lightweight descriptors for progressive loop detection with asynchronous ADMM pose-graph optimization [185]. Under simulated packet loss and latency, a two-stage Gauss–Seidel baseline fails to converge while Multi-Proxy continues, although the method relies on stable front-end odometry. Parallel graph inference separates current-state from historical-message inference [186], and CoVOR-SLAM shares poses, covariances, and ranges instead of features [76]. Swarm-SLAM runs decentralized inference with sparse loop prioritization online on a heterogeneous physical team [169], and foundation-model loop closing anchors monocular scale

TABLE VI
MULTI-ROBOT COORDINATION: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
ROAM [161]	One-hop consensus for semantic mapping and viewpoint planning. *Inputs:* RGB-D, LiDAR	Hardware, Sim. *Distributed mapping and exploration*	skid steer *Onboard*
Active graph exploration [162]†	Submodular loop-edge selection after coverage planning. *Inputs:* prior topo-metric graph	Sim. *Simulated coverage and graph objective*	none *Desktop evaluation*
Range integrity [163]†	Range-consistency localization-error reconstruction. *Inputs:* emulated ranges, reported poses	Online monitor (mixed reality), Sim. *Error reconstruction with emulated ranges*	aerial *Central ground station; emulated ranges*
Commerge [164]	Descriptor matching and scan selection before submap registration. *Inputs:* 3D LiDAR, LIO poses	Acquired data, Replay *Map-merging error and transmitted data*	wheeled (drive unspecified) *Desktop evaluation*
iMESA [165]†	Incremental local factor graphs with shared-variable consensus. *Inputs:* odometry and inter-robot factors	Replay, Numerical *Estimation error with limited communication*	mixed *Desktop evaluation*
Collaborative map repair [78]†	Residual-based fault isolation and correlation-aware map repair. *Inputs:* LiDAR, wheel odometry; GNSS initialization	Acquired data, Sim. *Pose/map error and fault isolation*	road vehicle *Unspecified*
TAPM-IVP [166]	Terrain-aware 3D Voronoi allocation and subgoal selection. *Inputs:* LiDAR, A-LOAM poses	Hardware, Sim. *Success, travel cost, and path duplication*	Ackermann *Onboard*
Failure-aware tracking [167]†	Failure-dependent tracking modes and target-state fusion. *Inputs:* target range/bearing and inter-robot messages	Hardware, Sim. *Target-state error during failures*	aerial *Unspecified*
Cross-rank [168]†	Intersection allocation from shared position histories. *Inputs:* LiDAR localization and low-bandwidth radio	Hardware, Sim. *Exploration under intermittent reception*	legged *Mixed placement*
CoVOR-SLAM [76]	Similarity-pose-graph fusion for monocular scale correction. *Inputs:* camera, inter-rover UWB ranges	Acquired data, Replay, Sim. *Pose error and monocular scale correction*	wheeled (drive unspecified) *Desktop evaluation*
Swarm-SLAM [169]	Sparse loop prioritization and decentralized inference. *Inputs:* local odometry, loop observations	Online estimate, Replay *Online collaborative estimation; team exchange*	skid steer *Onboard*
Planned synchronization [170]	Joint motion and observation scheduling. *Inputs:* relative marker observations, IMU	Hardware, Numerical *Formation and check-in cost; modeled non-idealities*	differential drive *Onboard*
Bandwidth-efficient SLAM [171]	Lossy keyframe selection and lossless coding. *Inputs:* visual/inertial keyframes	Online estimate, Replay *Bandwidth/accuracy tradeoff; central back end*	wheeled (drive unspecified) *Mixed placement*

† Road-vehicle, non-wheeled, handheld, or mixed-platform evaluations.

with metric odometry in Jetson replay [187]. Radio geometry adds constraints with their own assumptions, as in cooperative range-only SLAM analyzed offline with all landmarks visible [188] and formation-controlled UWB anchors whose geometry competes with obstacle clearance [189]. A prior-graph extension adds informative loop edges to multi-robot coverage paths [162], building on single-robot SLAM-aware planning [105], although its approximation bound concerns the graph objective rather than physical loop correspondences.

C. Rendezvous, disconnection, and conditional repair

Communication can become part of the motion objective. A rendezvous-aware explorer creates backtracking frontiers from decaying trajectory records [190], and a communication-constrained controller schedules sub-team meetings that require full-team agreement to update [191]. CoCoPlan co-optimizes task assignment with team-wise communication events through branch-and-bound search [192], and simulations with up to 100 robots report a 22.4% higher task-completion rate and 58.6% lower communication overhead than the compared protocols. MEF-Explore derives rendezvous from entropy fields while assuming that low-rate position exchange is always available [193], and three TurtleBot2 robots explore 21.32% faster with a 16.67% higher success rate than the baseline. All of these methods depend on reaching an encounter, exchanging the required state, and executing the subsequent plan.

During disconnection, robots can continue locally and reconcile later. ROAM continues planning from local maps during observed Wi-Fi cutoffs at the cost of revisits [161]. Correlation-aware collaboration exchanges states, covariances, and cross-covariances of common landmarks before

Kullback–Leibler averaging and Schmidt–Kalman updates, which excludes faulty observations and corrects uncertain landmarks [78]. Failure-aware target tracking fuses target estimates by covariance intersection after reconnection and lets a permanently isolated drone signal a hazard [167]. The communication-aware CBF controller relaxes conflicting constraints and drops unreliable links, with recovery evaluated under partial connectivity [20]. Planned synchronization makes observation an action, scheduling check-ins in which a TurtleBot pair stops and pans to observe each other [170]. A range-based integrity monitor reconstructs shared localization errors with distributed ADMM, although its demonstration runs centrally on emulated ranges [163]. These mechanisms address target estimates, landmarks, links, or observations, and none supplies a complete protocol for accepting stale or biased map content after an uncontrolled partition.

D. Motion interaction and persistent service

Coordination also occurs below the allocation layer. RL-RVO learns a distributed velocity policy from reciprocal velocity-obstacle features with external localization [194], predefined-time formation control compensates modeled actuator faults over a fixed graph [195], and compensation-based MPC estimates model discrepancy online for four motion-captured nonholonomic robots, leaving deadlock to future work [196]. Reachability-based platoon control preserves separation under bounded inputs while assuming reliable communication [197], and distributed barrier control adds a coordinated deadlock response on physical vehicles [198]. In mEclares, a scheduler reserves return and landing opportunities from energy and timing conditions, and agents fall back to committed paths after disconnection [199]. These results establish implemented coordination responses, whereas degraded-network validity requires its own fault model and trials.

E. Synthesis

Coordination methods reduce redundant work through allocation, compact representations, or distributed inference. Lower message volume changes which information is available without resolving inconsistencies between local estimates, and rendezvous planning, asynchronous optimization, and local planning sustain activity during disconnection. Reconnection additionally requires validating correspondences, accounting for correlations, and deciding which updates to accept. Evaluations with moving wheeled teams should therefore relate communication loss and reconnection to estimate divergence, coverage, interventions, and recovery time.

VIII. Fault Tolerance, Recovery, and Long-Term Deployment

Fault management includes detecting anomalies, isolating causes, accommodating degraded inputs or actuators, and restoring operation. These functions need not occur together, since a controller may tolerate bias without identifying its source and an estimator may correct a landmark without changing the motion of the robot. Table VII compares monitored quantities, response mechanisms, and evaluation settings.

A. Fault Estimation and Tolerant Control

Analytic redundancy provides diagnosis without duplicating every sensor. Kinematics-based detection compares position, angular velocity, and acceleration through transport relations, although its rover evaluation injects faults into recorded data whose substitute position source already contains inertial measurements [203]. An LPV vehicle diagnoser combines zonotopic set-membership estimation with invariant-set bounds, where separation from the healthy set establishes detectability and overlap between fault classes prevents unique isolation [201]. Tolerant controllers instead accommodate specified channels. Mixed-gain prescribed-performance control tolerates partial wheel-effectiveness loss [209], virtual-error control tolerates heading and wheel-speed faults while assuming exact position [210], adaptive sliding-mode control handles injected actuator faults and saturation on industrial hardware while requiring positive remaining actuation [211], and fixed-time adaptive tracking accommodates dead zones and modeled actuator faults in MATLAB studies [212]. An observer-based event-triggered controller reconstructs modeled faults before issuing updates in hardware-in-the-loop tests of a low-speed, no-slip vehicle [205]. Fault tolerance in these studies is a model-conditioned accommodation of a channel rather than diagnosis of an arbitrary physical failure or recovery of perception and navigation.

B. Integrity and Failure Monitoring

Monitors differ in the failure event they characterize. Reliable Monte Carlo localization (MCL) estimates a latent localization-success state alongside pose and triggers free-space global relocalization [202]. Particle-filter integrity bounds undetected lateral error under association-fault hypotheses [16], and EKF-SLAM integrity also accumulates landmark fault risk, although its recorded example crosses the threshold only after localization fails [17]. A prior-line-map protection level predicts estimation error, with operational alerting left for future work [204]. Learned-policy monitors use statistical alarm definitions, including conformal anomaly detection with nominal false-alarm control under exchangeability [213] and Sentinel, which pairs fast action-consistency monitoring with slower vision-language progress assessment [206]. PAAD predicts anomalies along a planned path on field recordings, while people reposition the robot after failures [214]. RADIUM moves failure prediction before deployment by sampling environment parameters from a distribution conditioned on failure [215]. It repairs the policy on the discovered cases and reports up to five times higher robustness for a repaired vision-in-the-loop policy on a 1/10-scale car, while failures absent from the simulator remain unpredictable. A latent reliability score, a bound on hazardous error, nominal false-alarm control, and simulation-based falsification answer different questions, and a useful warning must also leave time for an available response.

C. From Diagnosis to an Executed Response

The response must match the location of the fault in the stack. Switch-SLAM changes estimator constraints when one

TABLE VII
RELIABILITY AND RECOVERY: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
Visual NMPC recovery [200]	Tracking-loss stop and relocalization; restart after failure. *Inputs:* stereo vision, actuator feedback	Hardware *Tracking; stop/relocalization handling*	skid steer *Unspecified*
Set-based LPV diagnosis [201]†	Zonotopic detection and fault-class separation. *Inputs:* yaw rate, odometry, sideslip, steering angle	Acquired data *Detectability and isolability under injected faults*	road vehicle *Desktop evaluation*
Reliable MCL [202]	Joint pose/reliability estimation with global relocalization. *Inputs:* 2D LiDAR, odometry or inertial navigation	Acquired data, Replay, Sim. *Reliability estimates and relocalization*	wheeled (drive unspecified) *Unspecified*
Long-term home deployment [3]	Long-term service, docking, and remote human supervision. *Inputs:* LiDAR, RGB-D, docking and home sensors	Hardware *Service outcomes and interruptions*	wheeled (drive unspecified) *Mixed placement*
KSFD [203]	Kinematic residuals for sensor-fault diagnosis. *Inputs:* inertial channels and an inertial-aided position substitute	Acquired data, Numerical *Fault-isolation residual patterns*	Ackermann *Desktop evaluation*
Behavior-tree recovery [13]†	Behavior-tree supervision and failure-specific recovery. *Inputs:* terrain maps, contacts, planner status	Hardware, Sim. *Recovery on retained scenarios; human-rescue boundary*	mixed *Onboard*
Line-map integrity [204]	Prior-line-map localization with an error protection level. *Inputs:* monocular images, IMU, prior 3D lines	Acquired data, Replay, Sim. *Pose error and protection-level coverage*	wheeled (drive unspecified) *Desktop evaluation*
Event-triggered tracking [205]	Observer-based fault estimation and event-triggered tracking. *Inputs:* vehicle-posture outputs in the model	Hardware-in-the-loop, Numerical *Tracking error and update frequency*	Ackermann *Mixed placement*
Sentinel [206]	Action-consistency and VLM progress monitoring. *Inputs:* external RGB-D and mobile-base motion capture	Online estimate, Sim. *Failure detection and nominal alarms*	omnidirectional *Mixed placement*
RTEB [12]	Hybrid A* reinitialization and local smoothing. *Inputs:* RTK pose, voxel obstacle map	Hardware, Sim. *Planner recovery; TEB/NMPC comparisons*	wheeled (drive unspecified) *Onboard*
SCALE [207]	Affordance prediction and anti-novelty recovery. *Inputs:* RGB at deployment; offline robot data	Hardware *Visual recovery and intervention need*	skid steer *Onboard*
LiReN [208]	Online RL with resets, goals, and charging. *Inputs:* RGB policy; LiDAR/map, bumper for supervision	Hardware *Adaptation with operational resets and infrastructure*	differential drive *Mixed placement*

† Road-vehicle, non-wheeled, handheld, or mixed-platform evaluations.

modality fails [29], and fault-tolerant prior-map collaboration excludes faulty observations, inflates faulty-landmark covariance, and specifies reinitialization for odometry faults without evaluating an executed reset [78]. A GP predictor of safe-corridor optimizer failure triggers a stop–reposition–resume sequence [216]. With the same planner in simulation, recovery yields eight successes in ten runs versus three in ten in a dense world with 1 m doorways, whereas both variants succeed in all ten runs in a sparse world with 2 m doorways. The predictor concerns planner failure, and its posterior is not a collision probability. A fault-aware CBF controller optimizes over candidate sensor or actuator fault sets in numerical examples [217], and FORTRESS caches slow semantic reasoning for runtime reach-avoid fallback on a quadrotor [218].

Implemented recovery policies differ in what they restore. An NMPC visual navigation system stops after tracking loss and attempts relocalization, restarting the trial if it fails [200], and behavior-tree supervision on CENTAURO selects contact repositioning, reversing, replanning, or human assistance [13]. Resilient temporal-logic planning relaxes task windows or removes tasks when local correction fails [219], and predictive swarm fault tolerance withdraws an affected robot to maintenance before complete failure [220]. RTEB reinitializes an infeasible local trajectory with hybrid A* in a ROS 2 stack [12]. SCALE favors trajectories returning toward familiar visual states [207], whereas CARE applies repulsive corrections from estimated depth to a visual policy [221]. Neither familiarity nor estimated depth, however, supplies a clearance certificate. Help Me Through learns viewpoint actions from about 50 h of human demonstrations to keep monocular SLAM tracking through texture-poor transitions [222]. Its physical test runs the policy on live images while a person teleoperates path planning. When a robot cannot restore a capability, the team can reassign its task, as in energy-aware allocation under staged capability loss on Robotarium [223] and in R-FAC, whose log-sum-exp value factorization keeps search policies

effective when members fail without notification, assuming perfect target detection [224]. Task reports should state which requirements were completed, relaxed, transferred, or abandoned, and whether the robot could still execute the selected response.

D. Resources and Sustained Operation

Resource choices determine whether a response remains available. CODEI selects bodies, sensors, algorithms, and computing units from catalogs by coupling simulated planner queries to characterized perception errors [225]. Joint runtime control of motor speed and processor voltage and frequency on a wheeled rover reduces total energy by up to 36.34% relative to prior methods [226]. Its measurements show that computation can dominate the energy of slow motion. Fleet availability adds charging constraints, as in cyclic scheduling of operating and charging periods under a station budget [227] and in persistent-monitoring routes that minimize team size under revisit-latency and recharging limits with deterministic discharge [228]. A ten-home assistive deployment combines docking rules, cloud orchestration, and remote supervision, with internet problems and localization failures among the causes of interruption [3]. A cloud–edge system offloads mapping and trajectory generation to a Kubernetes cluster over 5G while keeping MPC tracking and collision avoidance on Jetson-equipped Ackermann robots, with connection modes that fall back to a leader-based local network [229].

Maintenance and learning form a second operational loop. Predictive data acquisition selects new training observations for teach-repeat-learn navigation from visual matching ambiguity [230]. SELFI stabilizes deployment-time RL with a model-based objective and counts interventions during learning [231], LiReN combines foundation-model adaptation with pseudo-resets, keepout rules, charging, and help escalation [208], and counterfactual replay selects among adapting policies at the cost of between-trial computation [232]. An outdoor visual-inertial benchmark shows why failure and resource reporting must remain coupled, since loop closure changes both accuracy and compute while failed and low-coverage runs are removed before averaging [233].

E. Synthesis

Fault tolerance is specific to the modeled channel. Remaining actuation sustains tracking, redundant observations support diagnosis, and an alternative local state permits planner recovery, while a monitor is useful only if its failure event matches the response being selected. Long-term service additionally depends on charging, energy, network access, offboard computation, and human assistance, and recovery quality therefore includes both restored operation and the interruptions needed to achieve it.

IX. Evaluation Practice

Evaluation of autonomy under degradation must relate component outputs to executed behavior. Three axes organize the comparison: the level at which performance is measured, the conditions under which degradation is tested, and the event and assumptions behind a risk or diagnostic claim. Fig. 3 connects the axes through one time-aligned episode, and Table VIII counts the attributes reported by the 228 technical publications. These counts describe the selected sample rather than the field.

TABLE VIII
Reported evaluation attributes among the 228 technical publications. Attributes overlap; context publications are excluded.

Attribute	Publications
Measurement level	
Component measures	160
Task outcomes	128
Test conditions	
Degradation testing	130
Risk and diagnostic assessment	
Integrity/reliability	14

A. Measurement, Test Conditions, and Claim Interpretation

Component measures include trajectory, mapping, segmentation, prediction, and tracking error together with runtime and classification scores, whereas *task outcomes* concern completion, collision, coverage, traversal time, intervention, and service availability. Lower error can help explain a task result, although the connection must be measured rather than assumed. *Degradation testing* covers injected faults, outages, terrain variation, and deliberately selected difficult sequences, and it describes an experimental condition rather than a metric. For *risk and diagnostic assessment*, the defined event matters, since classifier recall, nominal false-positive control, protection-level coverage, and undetected hazardous error are not interchangeable, as reliable MCL, conformal anomaly detection, and EKF-SLAM integrity illustrate [17], [202], [213].

B. Execution and Reference Dependence

Recorded measurements preserve sensor artifacts without testing feedback between the estimated state and future observations, whereas closed-loop evaluation lets the evaluated output change the commands and thus the subsequent observations. External pose, offboard processing, teleoperated prior mapping, and staged scenes define the conditions of that feedback. Slip-aware fusion is evaluated during physical motion without a controller consuming its estimates [46], LiDAR-based barrier control runs under external pose and offboard computation [15], and a learned clearance model plans on a laptop connected over a local network [122]. The step to hardware can itself change the result, as when terrain-adaptive MPC falls from 94% success in simulation to 80% outdoors [99].

References introduce their own dependence when they share sensors or processing with the evaluated method. Campus line localization uses a map and trajectory reference from the same LiDAR–inertial pipeline [204], kinematic diagnosis

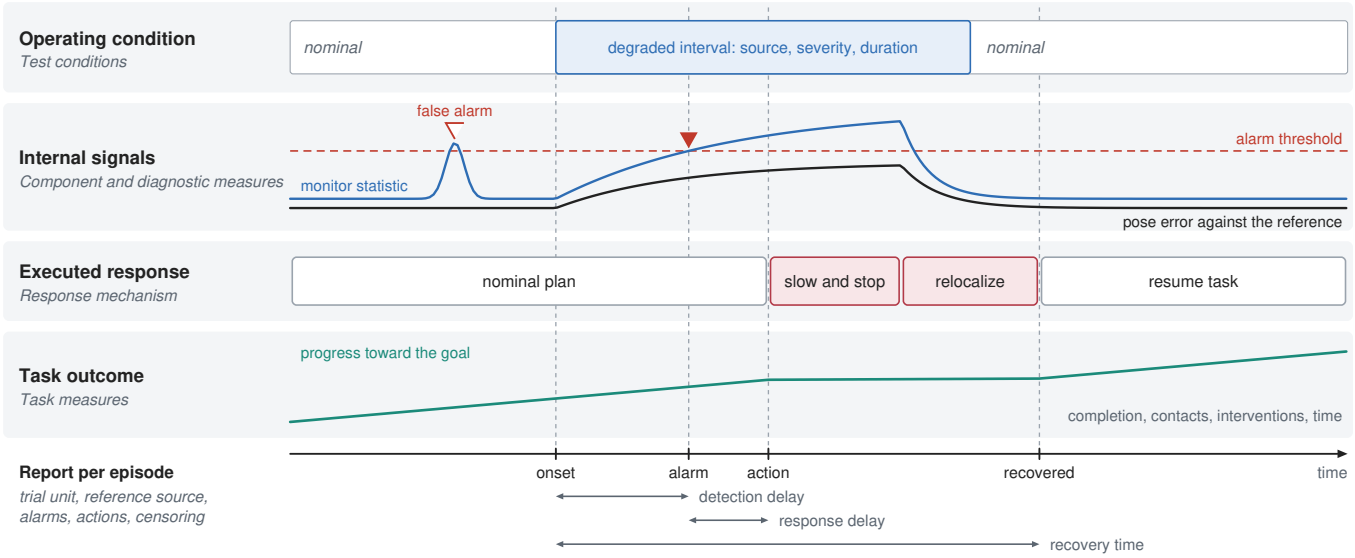


Fig. 3. Joint evaluation of detection and response in one navigation episode. The operating condition defines the test, internal signals carry component and diagnostic measures, and the executed response and task outcome carry task measures. Detection delay, response delay, and recovery time follow from the synchronized events, and a threshold crossing without degradation counts as a false alarm.

uses a substitute position that shares sensors with the monitored signals [203], and M2UD mixes smoothed RTK, offline multisensor SLAM, and selected laser-tracker checks [2]. The HK-MEMS bus-route reference lacks timestamps [19], GEODE varies reference construction by scene [18], and orchard map merging is scored against the same mapping backbone [183]. Cooperative range-only SLAM runs offline on joystick-recorded bags [188], and hierarchical map transmission reports an estimated risk field rather than measured contacts [182]. Controlled variables must also match the claimed mechanism. Varying terrain geometry tests plane-fitting allocation [166], whereas a test of pose-aware terrain decisions must vary pose uncertainty. Similarly, URGE ranks methods under ideal simulated communication [178], and riMESA replays benchmark factors with modeled connectivity [184], which means that neither setting measures how physical link loss changes the ranking.

C. Guarantees, Rates, and Denominators

A formal safety result applies to a specified event, probability space, horizon, and set of assumptions. A scenario-planning certificate concerns the risk of an accepted plan under a predicted obstacle distribution [90], environment-level perception calibration depends on representative calibration environments and known state [92], and a barrier proof may assume feasible inputs that saturation violates [15]. Spatiotemporal-tube guarantees likewise assume terminal separation without explicit input limits [91]. Empirical closed-loop collision frequency is a different quantity.

Empirical rates require a denominator and an independent trial unit. Across ten homes, the assistive-robot study records 1539 service requests, of which 1130 succeeded, 177 failed, and 232 were not performed [3]. Its 86.46% success rate uses the 1307 attempted interventions as denominator and is therefore not the fraction of requests served. Eligibility and censoring change the represented population. The CENTAURO study discards infeasible or slowly planned scenarios before reporting outcomes [13], cross-modal teach-and-repeat excludes methods that fail on the first route from later routes [75], an outdoor visual-inertial benchmark averages error only over surviving runs [233], and multi-session radar mapping omits failed baselines from its averages [61]. HK-MEMS also warns that a one-sided distance score can hide a shortened trajectory [19]. A trial repeated after failed relocalization [200] differs from recovery within the original attempt, and completion after temporal-logic relaxation [219] must name the changed requirements. Small hardware samples, such as 30 object-search trials [144] or few repetitions per outdoor route [154], support feasibility more than precise rates. The evaluated event matters as well, since a route choice around a virtual risky branch [95] is not a measured damage reduction, conformal coverage and intrusion limits [121] are not mission collision rates, and learned terrain and semantic uncertainty [32], [81] still requires calibration. Safety and completion must also be reported together, since refusing every action reduces collisions while preventing useful work.

D. Timing, Placement, and Comparisons

Timing should cover sensing age, transport, perception, estimation, planning, supervision, command delivery, and actuator response, which processor identity and average module runtime do not establish. Sentinel's fast and slow monitors have different intervention horizons [206], FORTRESS caches slow reasoning before runtime planning [218], and a 1 Hz VLM planner [155] or a change detector at 2.74 Hz on an embedded GPU [60] requires a local controller that stays valid between outputs. The compute-sensitive trajectory metric of M2UD supports estimator comparison without measuring control-loop delay [2]. DMSA processes data at 1.5–4 times the recording duration and distinguishes unsupported sensor

adaptations from failed runs [28], and memory-event-triggered MPC counts updates on an industrial computer rather than measuring onboard execution [128]. A study of LiDAR–inertial non-reproducibility repeats each experiment 30 times after controlling subscribers and buffering [234], showing that repeated playback characterizes implementation variability rather than variation across missions. Comparisons with simulator pose characterize planning, whereas ablations with a SLAM back end expose the effects of estimated pose [87]. Adaptation, preparation, and energy belong in the same accounting, including workstation training and server updates for deployed learners [208], [231], between-trial replay [232], map or tour preparation and cloud calls for language navigation [140], [151], [152], and the computational share of motion energy [226].

E. Joint Evaluation of Detection and Response

A joint evaluation links degradation onset to an internal trigger, an executed response, and a task outcome within one mission, as Fig. 3 shows. Its specification includes the fault model, onset, duration, severity, recovery criterion, and channels assumed to remain trustworthy. Synchronized logs should retain detector outputs, thresholds, false and missed alarms, detection-to-action delay, and which of observation rejection, map correction, filter reset, or physical maneuver was executed. Comparing the nominal stack, a simple response, and an integrated response under matched conditions isolates the contribution of the response mechanism, while no-fault controls reveal the cost of unnecessary interventions. Communication studies should separate intermittent reception, complete partition, stale messages, and reconnection, with conditional map repair as one baseline [78], and semantic systems such as GSON [139] should be tested with delayed, unavailable, and unsafe model responses. Outcomes should combine completion, contact, interventions, path and time cost, and resource use with component error and diagnostic calibration, reported with the trial unit, censoring, reference coverage, and uncertainty intervals where estimable.

X. Open Challenges

The reviewed systems already couple estimation with exploration, semantic reasoning with geometric control, and failure prediction with recovery. The four interfaces below, numbered as in Figs. 1 and 2, connect these functions in ways whose behavior under degradation remains insufficiently characterized. For each interface, the discussion identifies the existing mechanisms, the unresolved dependency, and a matched comparison that would resolve it under the evaluation principles of Section IX.

A. Localization-Triggered Navigation Recovery

Localization monitoring and executed recovery are both available, although the reviewed systems rarely connect them through the same trigger. Protection levels characterize pose-error coverage [204], modality switching restores estimator constraints [29], collaborative repair revises landmarks [78], and fault-tolerant tracking accommodates specified channels [205]. On the action side, GP-based recovery executes a stop–reposition–resume sequence after optimizer failure [216], visual tracking loss triggers a stop and a relocalization attempt on a physical vehicle [200], and CENTAURO selects failure-specific recovery through behavior-tree supervision [13]. The vehicle result can require a restarted trial, whereas CENTAURO relies on hybrid motion primitives and can require human release. RTEB repairs infeasible plan initialization, SCALE returns toward familiar observations, and CARE adjusts trajectories using estimated depth [12], [207], [221]. Active view planning tries to prevent visual tracking loss before it occurs, although its physical test uses teleoperated motion [222]. Wheel–inertial odometry with a ground-manifold prior offers an interoceptive fallback whose drift grows without external correction [45]. These triggers concern different objects, namely optimizer infeasibility, visual novelty, predicted tracking loss, and hazardous pose error. The open question is how a localization warning should select slowing, stopping, relocalization, or repositioning when part of the state needed to execute that action is itself unreliable.

A discriminating comparison would keep the estimator and route fixed while varying only the response, namely an immediate stop, threshold-triggered relocalization, or an action selected from the remaining trustworthy channels. Controlled pose bias and weak geometry would separate incorrect-state faults from simple measurement loss. Trials in which one alarm fires while another channel is misleading would test the supervisor rather than the recovery maneuver alone. The decisive outcome is whether the selected response restores localization and task progress without increasing contact or intervention, with the cost of unnecessary stops under nominal sensing reported alongside. For compact conventional wheels, the candidate actions must also remain feasible with the sensing and actuation left after the fault.

B. Resource-Constrained Semantic Reasoning and Fallback

Wheeled systems already combine semantic trajectory selection with geometric planning [138], [154], group reasoning with constrained local control [139], and embodiment-specific re-ranking that rejects paths the chassis cannot execute [155]. Perception calibration [92], runtime monitoring [206], cached reasoning with fallback [218], and distillation into lightweight encoders [158] address reliability and response time. Language-derived costmap layers [157] and relocation-aware object search [144] move semantic errors into costs or search order, where geometry can check them. The remaining interface concerns the lifetime of a semantic decision, since an old destination can remain valid while an old pedestrian detour becomes unusable. In the team-level analogue, Co-NavGPT executes semantic frontier assignments geometrically with perception and planning on a workstation [132]. A delayed assignment can then remain syntactically valid after the shared map or the teammate allocation has changed. HAM-Nav and Mobility VLA depend on sketch or tour coverage, while VLM-GroNav retains GPS and cloud dependencies [141], [151], [152]. SocialNav-SUB indicates

that incorrect reasoning, and not only unavailable or delayed responses, requires explicit testing [142].

A matched comparison could hold one model and controller fixed while comparing periodic querying, scene-triggered querying, and cached decisions. Delayed responses combined with changed candidate paths would reveal whether stale-result rejection improves navigation beyond reducing query frequency. Injected incorrect responses would test whether a fast but wrong social judgment is rejected. Completion, contact, and fallback duration should be weighed against energy and query cost. Varying obstacle-track quality would separate failures of semantic scheduling from failures of the state used by local constraints, since local geometric control preserves motion only while its sensing and feasible-action assumptions hold. For teams, assignment age and rejection should be evaluated alongside local fallback.

C. Pose Uncertainty in Wheeled Terrain Decisions

Recent terrain decisions increasingly combine geometry with localization, map, or contact uncertainty. Terrain-aware Ackermann collaboration uses local geometry and spatial allocation [166], FIT-SLAM 2 couples traversability with expected localizability [87], and an aquatic explorer weights mapping information by positional uncertainty [82]. Semantic neural processes estimate uncertain elevation [32], evidential semantic maps attach uncertainty to terrain classes [81], visual-geometric maps use novelty and speed distributions in executed navigation [94], and SPARTA models angle-dependent risk [95]. Terrain-adaptive stability bounds [99] and implicit traversability fields [100] shape executed speed and route on small skid-steer robots, whose hardware failures are attributed to wheel slip, wet-ground traction, and sensor occlusion rather than to pose error. Terrain-point scatter, expected landmark information, and pose covariance thus describe different quantities. The open question is how uncertainty in pose alters the probability that a wheel-supported footprint lies on the estimated traversable surface under incomplete observation.

A useful comparison would propagate a calibrated pose distribution into footprint clearance or speed while keeping the terrain map and chassis fixed, with geometry-only planning and fixed inflation as baselines. Narrow support regions and slopes would expose the effect of translational and heading uncertainty on wheel placement. The result should show whether propagation reduces contact or intervention at an acceptable cost in completion and travel time, with aborted risky attempts counted as outcomes. Independent reference coverage is needed to distinguish true pose uncertainty from shared estimation error. Datasets such as M2UD support estimator comparisons on difficult ground-robot sequences [2], whereas action-dependent terrain decisions require closed-loop trials that also distinguish uncertain obstacle location from uncertain traversal consequence.

D. Disconnection, Reconnection, and Corrupted Shared Information

Reconnection mechanisms exist for particular shared quantities, such as the target estimates that failure-aware tracking fuses after a link returns [167]. Cooperative pose fusion [76] and correlation-aware landmark repair [78] address other estimation objects. Reception-aware control [20], sparse position sharing [168], and rendezvous planning co-optimized with task allocation [192] change connectivity or message requirements. Asynchronous consensus back ends tolerate dropped exchanges and outlier measurements [184], [185], whereas regional maps converge through a deterministic merge rule that tolerates delay without testing corrupted content [178]. Sparse loop exchange, lossy keyframe selection with lossless coding, layered transmission with back-end feedback, and planned observation check-ins reduce traffic [169]–[171], [182], although lower traffic does not establish safe acceptance after reconnection. Range-consistency monitoring reconstructs shared localization errors under a centralized mixed-reality protocol [163], whereas a moving wheeled team adds physical range corruption and limited communication. The open interface is selective acceptance when isolated robots return with stale, correlated, or biased information, since receiving a packet establishes neither frame compatibility nor the validity of its map or pose content.

Trials with moving ground robots could compare immediate fusion, age-based rejection, and correlation- or residual-aware acceptance after a controlled partition, with the deterministic merge as a baseline. Delay and local landmark bias should be varied separately, since an old estimate need not be faulty and a recent one need not be correct. Fault propagation and shared-state divergence would establish the estimator effect, whereas duplicated coverage, completion, and interventions in revisitable environments would establish the coordination consequence. Stable metric odometry in foundation-model loop fusion [187], continuous range visibility in cooperative initialization [188], orchard-specific overlap priors [183], and stable front-end odometry in asynchronous collaborative LiDAR SLAM [185] are concrete assumptions to violate one at a time. Such trials would reveal whether the loss of a prerequisite is detected before shared information changes team motion.

E. Broader Challenges

1) Standardized multifactor benchmarks: Degradation sources interact, as when poor illumination coincides with moving obstacles and weak network coverage. Benchmarks therefore need controlled severity and onset, factorial or stratified designs, repeated independent missions, and annotated reference quality, with module scores linked to completion, contacts, and interventions. Ground-robot datasets [1], [2] support reproducible estimator comparisons, whereas action-dependent sensor feedback requires closed-loop evaluation.

2) Transfer from simulation and prior experience: Terrain-policy failures [113], the supervision requirements of learned wheel models [47], and a terrain field trained on one skid-steer platform and deployed on a lighter one [100] motivate comparisons between explicit model adaptation and training-data variation. Simulation-based failure prediction can repair policies before deployment, although only for the failures that the simulator models [215]. Evaluations should report failure

onset, recovery, additional data requirements, and performance on withheld terrain, sensor, and layout conditions.

3) Computation, energy, and information age: Resource-aware design [225], cached semantic reasoning [218], cloud–edge offloading [229], and joint speed and processor control [226] connect computation choices to navigation. Response-time distributions under concurrent sensing, mapping, and reasoning loads are needed to quantify deadline misses and stale-result rejection. Energy per completed task would link computational savings to mission utility, since isolated execution-time gains may be offset by information delays elsewhere in the system.

4) Long-term map and service maintenance: Persistent mapping [27], change detection on prior maps [60], cross-modal teach-and-repeat over months [75], dynamic scene estimation [55], and home deployment [3] require representations to be revised while localization and service continue. Longitudinal evaluations should relate map updates to revisits, storage growth, inconsistencies, availability, maintenance effort, and human assistance, distinguishing the time between recording sessions from the duration of uninterrupted autonomy. Seasonal localization maps and simultaneous delayed-overlap merging also remain distinct temporal claims [67], [183].

5) Assurance under changing information quality: Integrity analysis [17], statistical perception assurance [92], risk-aware planning [90], and learned clearance distributions [122] rest on different assumptions about faults, distributions, references, and feasible actions, any of which may cease to hold during operation. Online monitors of these assumptions should be evaluated under distribution shift, actuator saturation, infeasible constraints, and sensing delays, with false-alarm and missed-violation rates reported together with the consequences of the subsequent response. Continual learning further depends on the infrastructure that makes new data useful. SELFI, LiReN, and OLiVia-Nav combine adaptation with localization, resets, supervisory support, or online encoder updates [158], [208], [231]. Longitudinal comparisons of such systems should retain failed learning attempts, charging and network downtime, map preparation, and human escalation alongside post-adaptation success.

6) Generalization across embodiments: Differential drive, skid steering, and Ackermann steering impose different curvature, traction, and tracking constraints. Swapping an embodiment-specific affordance model has let one semantic planner serve legged and wheeled robots [155], while failure-aware team policies keep a search effective when members drop out [224]. Cross-platform evaluations should report clearance, curvature feasibility, traction, payload disturbance, and timing margins alongside success. Results obtained with high-resolution sensing or external pose must also be reassessed under the sensing and computation of the target robot.

XI. Conclusion

This survey compared methods for wheeled autonomy through the navigation assumptions affected by degradation, the information or actions changed by each response, and the evidence for their effectiveness. Estimators retain useful constraints through selective updates, gain independent motion information from radar, range, and wheel sensing, and extend motion prediction through model adaptation, while navigation systems revise maps to correct accumulated inconsistency. These responses depend on different operating conditions, namely surviving geometry, credible supervision, valid correspondences, and consistent correlations, none of which can substitute for another.

The reviewed results also reveal conditional trade-offs. More observable exploration routes can cost additional travel and computation, risk-constrained planning trades conservatism against progress, and recovery helps when its predicted failure matches the obstacle to task execution. Semantic ranking remains limited by the available candidates and by the capabilities of the executing chassis, whereas compact or asynchronous communication preserves only the information encoded in the shared representation. These dependencies explain why improved component scores alone do not establish more reliable navigation.

The central requirement for further evaluation is to connect degradation onset, an internal trigger, and an executed response within the same mission, as the four interfaces of Section X illustrate. Comparisons that retain the trial unit, reference dependence, timing, and failure conditions can establish when an integrated response improves both task performance and safety. The heterogeneous, purposively selected evidence supports these mechanism-level conclusions, whereas broader performance rankings require matched protocols and more uniform coverage.

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